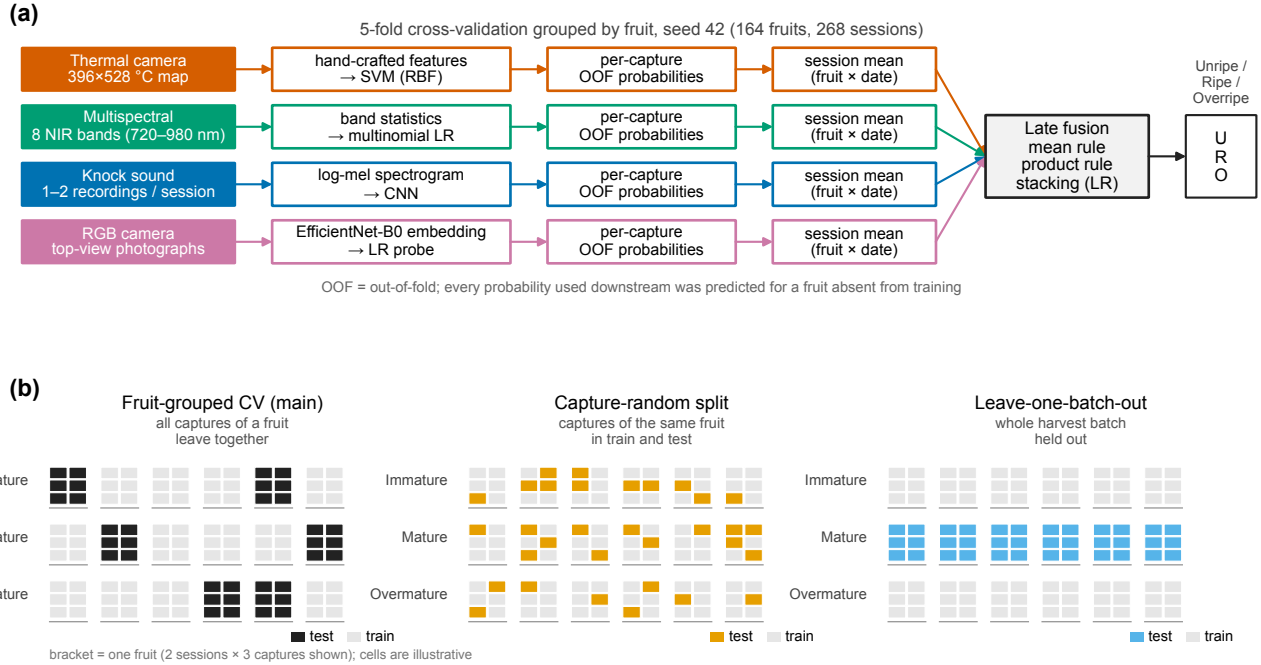
(a)
5-fold cross-validation grouped by fruit, seed 42 (164 fruits, 268 sessions)
Thermal camera
396×528 °C map
hand-crafted features
→ SVM (RBF)
per-capture
OOF probabilities
session mean
(fruit × date)
Multispectral
8 NIR bands (720–980 nm)
band statistics
→ multinomial LR
per-capture
OOF probabilities
session mean
(fruit × date)
Knock sound
1–2 recordings / session
log-mel spectrogram
→ CNN
per-capture
OOF probabilities
session mean
(fruit × date)
RGB camera
top-view photographs
EfficientNet-B0 embedding
→ LR probe
per-capture
OOF probabilities
session mean
(fruit × date)
Late fusion
mean rule
product rule
stacking (LR)
Unripe /
Ripe /
Overripe
U
R
O
OOF = out-of-fold; every probability used downstream was predicted for a fruit absent from training
(b)
Fruit-grouped CV (main)
all captures of a fruit
leave together
Capture-random split
captures of the same fruit
in train and test
Leave-one-batch-out
whole harvest batch
held out
Immature
Mature
Overmature
test
train
bracket = one fruit (2 sessions × 3 captures shown); cells are illustrative